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Effects of a combined non-dominant-limb-focused technical and unstable-surface balance training program on dynamic postural control, inter-limb asymmetry, and soccer-specific performance in young soccer players

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Background: Soccer involves frequent unilateral and bilateral actions that may contribute to differences between dominant and non-dominant limbs. Understanding whether targeted training strategies can modify limb-specific performance and inter-limb asymmetry patterns may provide valuable insights into bilateral adaptations in young soccer players.
Objective: This study aimed to investigate the effects of an 8-week combined training program incorporating non-dominant-leg-focused technical exercises and unstable-surface training on limb-specific dynamic postural control, inter-limb asymmetry, and soccer-specific performance in young soccer players.
Methods: Thirty male U12 soccer players were randomly assigned to an experimental group (n = 15) or a control group (n = 15). Both groups trained twice weekly for 8 weeks with the same training frequency and session duration. The experimental group performed three technical exercises focused on the non-dominant leg and one dynamic bipodal unstable-surface exercise performed on unstable balance cushions, whereas the control group followed conventional soccer training.
Results: Compared with conventional soccer training, the combined training program was associated with greater improvements in dynamic postural control and selected non-dominant-leg soccer-specific performance outcomes. Limb-specific analyses indicated that training-related adaptations differed according to limb condition and task characteristics. Improvements were also observed in selected absolute and signed asymmetry measures; however, asymmetry responses were not uniform across outcomes.

Conclusion: The findings suggest that an 8-week combined training program integrating non-dominant-leg technical exercises with dynamic bipodal unstable-surface exercises may improve selected aspects of dynamic postural control and soccer-specific performance in male U12 soccer players. Limb-specific analyses and signed asymmetry assessment provided additional information regarding bilateral adaptations. However, because the intervention combined two training components, the independent contribution of each component cannot be determined, and these findings should be considered preliminary.

KEYWORDS

bilateral coordination, motor development, neuromuscular adaptation, skill index, star excursion balance test, technical skills, youth athletes

1 Introduction

Soccer performance in children is shaped by the interaction of technical skill, neuromuscular function, and dynamic postural control (Read et al., 2019). Because key soccer actions such as passing, dribbling, ball control, shooting, and changes of direction are typically performed with a preferred limb, young players often develop functional asymmetries between the dominant and non-dominant legs (Bishop et al., 2018). From a practical perspective, improving the technical proficiency of the non-preferred foot has been shown to reduce lower-limb functional asymmetry in young soccer players (Guilherme et al., 2015). However, although inter-limb asymmetries have frequently been associated with impaired athletic performance, this relationship appears to be task-specific rather than universal, and excessive asymmetry should not necessarily be considered detrimental in all situations (Fox et al., 2023). Consequently, interventions aiming to improve bilateral competence should focus on enhancing functional performance rather than solely minimizing asymmetry.

Dynamic postural control is another key determinant of soccer performance because it contributes to lower-limb stability during kicking, landing, cutting, dribbling, and rapid changes of direction (Paillard, 2019). The Star Excursion Balance Test (SEBT) is widely recognized as a valid and practical tool for assessing dynamic postural control through reach performance and identifying side-to-side deficits in physically active populations (Gribble et al., 2012). Dynamic postural control represents one functional component of soccer performance, particularly during unilateral support phases such as kicking, changing direction, and landing. The Star Excursion Balance Test (SEBT) assesses multidirectional dynamic postural control under unilateral conditions; however, its interpretation should remain specific to functional reach performance and should not be considered a comprehensive assessment of all mechanisms involved in postural control. Importantly, neuromuscular interventions have demonstrated the capacity to improve dynamic postural control in young soccer players. For example, Jlid et al. (2019) reported that an 8-week multidirectional plyometric training program significantly improved SEBT performance, highlighting the trainability of balance-related neuromuscular function during youth development.

From an applied perspective, improving the non-dominant leg is particularly important because greater bilateral proficiency may enhance technical versatility and reduce excessive reliance on the preferred limb during match play (Carey et al., 2001). Previous studies have shown that specific non-preferred-foot practice can increase non-preferred-foot use while reducing functional asymmetry (Guilherme et al., 2015). Likewise, balance-oriented interventions have been shown to improve balance performance and selected technical skills in adolescent soccer players (Mitrousis et al., 2023). One potential mechanism is that unstable-surface balance training enhances proprioceptive acuity, sensorimotor integration, postural regulation, ankle–knee–hip coordination, trunk stability, and neuromuscular control under unstable conditions. Improvements in foot stability and ankle mobility may further contribute to more efficient movement control and landing mechanics, both of which are important for functional performance (Patti et al., 2024). Collectively, these neuromuscular adaptations may provide a foundation for improved dynamic postural control and facilitate the execution of soccer-specific technical skills.

Although soccer-specific actions are predominantly unilateral and performed under dynamic conditions, the balance exercises used in the present study were performed in a bipodal stance on unstable balance cushions. These exercises were selected because U12 players are at a developmental stage characterized by considerable neuromuscular plasticity, during which improvements in general postural regulation, sensorimotor integration, and neuromuscular control may contribute to more efficient movement execution during sport-specific tasks. However, adaptations resulting from bipodal unstable-surface balance exercises should not be interpreted as a direct transfer mechanism to unilateral soccer-specific actions, as unilateral and bilateral tasks involve different coordination constraints and task-specific motor-control demands. Therefore, unstable-surface balance training was intended to provide complementary postural-control and neuromuscular stimuli rather than directly reproduce or modify the coordination patterns required for non-dominant-limb technical skills.

Soccer performance requires repeated unilateral and bilateral actions involving complex interactions between technical execution, neuromuscular coordination, and postural regulation. Although inter-limb asymmetries in soccer have been extensively investigated, many training interventions have primarily focused on isolated components of performance, such as technical skill development or neuromuscular and balance training. The rationale for combining these approaches is not based solely on the limited evidence regarding their simultaneous application, but rather on their potential complementarity. Training strategies emphasizing non-dominant-limb technical involvement may

address skill-related differences associated with preferential limb use, whereas unstable-surface balance training may provide additional sensorimotor challenges that contribute to dynamic postural control and neuromuscular regulation. Therefore, a combined approach may provide a theoretically justified strategy to target multiple determinants of functional performance in young soccer players.

Despite this theoretical complementarity, evidence regarding the combined effects of non-dominant-leg technical exercises and unstable-surface balance training remains limited. Furthermore, although these two training modalities may have complementary mechanisms, the present two-arm randomized controlled design does not allow the independent contribution of each intervention component to be determined. Consequently, the present study evaluates the effectiveness of the combined intervention compared with conventional soccer training rather than the isolated effects of either training modality.

Therefore, the primary aim of the present study was to examine the effects of an 8-week combined training program integrating non-dominant-leg technical exercises with dynamic bipodal balance exercises performed on unstable balance cushions, compared with conventional soccer training, on dynamic postural control, assessed using the Star Excursion Balance Test (SEBT), in male U12 soccer players. Secondary outcomes included balance asymmetry, soccer-specific performance performed with both legs, skill index, inter-limb difference, and asymmetry percentage. We hypothesized that the experimental group would demonstrate greater improvements in dynamic postural control together with enhanced soccer-specific performance and reduced inter-limb asymmetry compared with the control group.

2 Methods

2.1 Participants

The participants were 30 male U12 outfield soccer players recruited from the same academy-level soccer program. Goalkeepers were excluded, and only outfield players were included in the study. All participants followed the same soccer training schedule before and during the study period, consisting of two training sessions per week. Baseline participant characteristics, including chronological age, maturity offset expressed as years from predicted peak height velocity, anthropometric characteristics, training age/soccer playing experience, playing-position distribution, and competitive level, are presented in Table 1. Participants were recruited, enrolled, and assessed for eligibility by a member of the research team. They were randomly allocated in a 1:1 ratio to an experimental group (n = 15) and a control group (n = 15) using a computer-generated randomization sequence prepared by an independent researcher who was not involved in participant recruitment, intervention delivery, or outcome assessment. Allocation concealment was ensured using sequentially numbered, opaque, sealed envelopes prepared by the independent researcher. After completion of the baseline assessments, the independent researcher opened the envelopes in numerical order and assigned participants to their respective groups. Leg dominance was determined according to Van Melick et al. (2017) and defined as the preferred kicking leg during soccer-specific actions. The limb habitually used for kicking was considered the dominant leg, whereas the contralateral limb was considered the non-dominant leg. All participants reported no recent injuries to the hip, knee, or ankle and no existing musculoskeletal disorders affecting the lower limbs or lower back.

2.2 Experimental design

This study was designed as a two-arm, parallel-group randomized controlled trial with a 1:1 allocation ratio and was reported in accordance with the CONSORT 2010 statement (Schulz et al., 2010). The study protocol was approved by the Ethics Committee of the Deanship of Scientific Research at Palestine Technical University–Kadoorie. The trial was retrospectively registered in ClinicalTrials.gov under the identifier NCT07853638. Before participation, all players and their parents or legal guardians received verbal and written information regarding the study procedures, potential risks, and benefits. Written informed consent was obtained from the parents or legal guardians of all participants in accordance with the Declaration of Helsinki. Participants were also

TABLE 1 Baseline characteristics of participants.

Variable	Control group (n = 15)	Experimental group (n = 15)	p-value
Age (years)	10.5 ± 0.7	10.8 ± 0.7	0.313
Maturity offset (years from predicted peak height velocity)	-2.6 ± 0.6	-2.4 ± 0.5	0.374
Height (m)	1.43 ± 0.07	1.44 ± 0.07	0.573
Body mass (kg)	40.7 ± 7.1	40.9 ± 8.2	0.925
Lower limb length (cm)	84.8 ± 3.2	85.4 ± 3.2	0.611
Body mass index (kg/m^2)	20.0 ± 3.8	19.6 ± 3.0	0.715
Training age / soccer practice experience (years)	2.6 ± 0.7	2.8 ± 0.7	0.445
Weekly training frequency (sessions/week)	2	2	—
Competitive level	Academy level	Academy level	—
Defenders, n	6	6	—
Midfielders, n	5	5	—
Forwards, n	4	4	—

informed that they could withdraw from the study at any time without consequence.

Thirty male U12 soccer players were assessed for eligibility, and all participants fulfilled the inclusion criteria. No participants were excluded before randomization. Following random allocation, participants were assigned to the experimental group (n = 15) or the control group (n = 15). The study was conducted using a randomized controlled pre–post experimental design. Participants were randomly assigned to an experimental group or a control group. The primary analysis was based on a 2 × 2 mixed design, with group (experimental vs control) as a between-subject factor and time (pre-intervention vs post-intervention) as a within-subject factor. For outcomes assessed separately in dominant and non-dominant limbs, additional 2 × 2 × 2 mixed analyses were conducted, including group, time, and limb as factors. All randomized participants completed the 8-week intervention, with no losses to follow-up, dropouts, exclusions from analysis, or missing outcome data. Therefore, all 30 randomized participants were included in the final analysis. Participant progression through eligibility assessment, randomization, allocation, follow-up, and final analysis is presented in the CONSORT flow diagram (Figure 1).

The intervention lasted eight weeks and was carried out during August and September. During this period, the experimental group completed a specific training program targeting the non-dominant leg and including dynamic bipodal postural balance exercises performed on cushions, whereas the control group continued their regular soccer training routine without any additional targeted intervention (Tables 2, 3). All performance assessments were conducted by the same assessor to ensure consistency in the testing procedures. Due to the nature of the training intervention, blinding of participants and coaches was not feasible. However, the assessor responsible for conducting the pre- and post-intervention performance evaluations was blinded to group allocation and was not involved in intervention delivery. During the assessment procedures, participants were identified using numerical codes, and the assessor was provided only with these identification numbers and was not

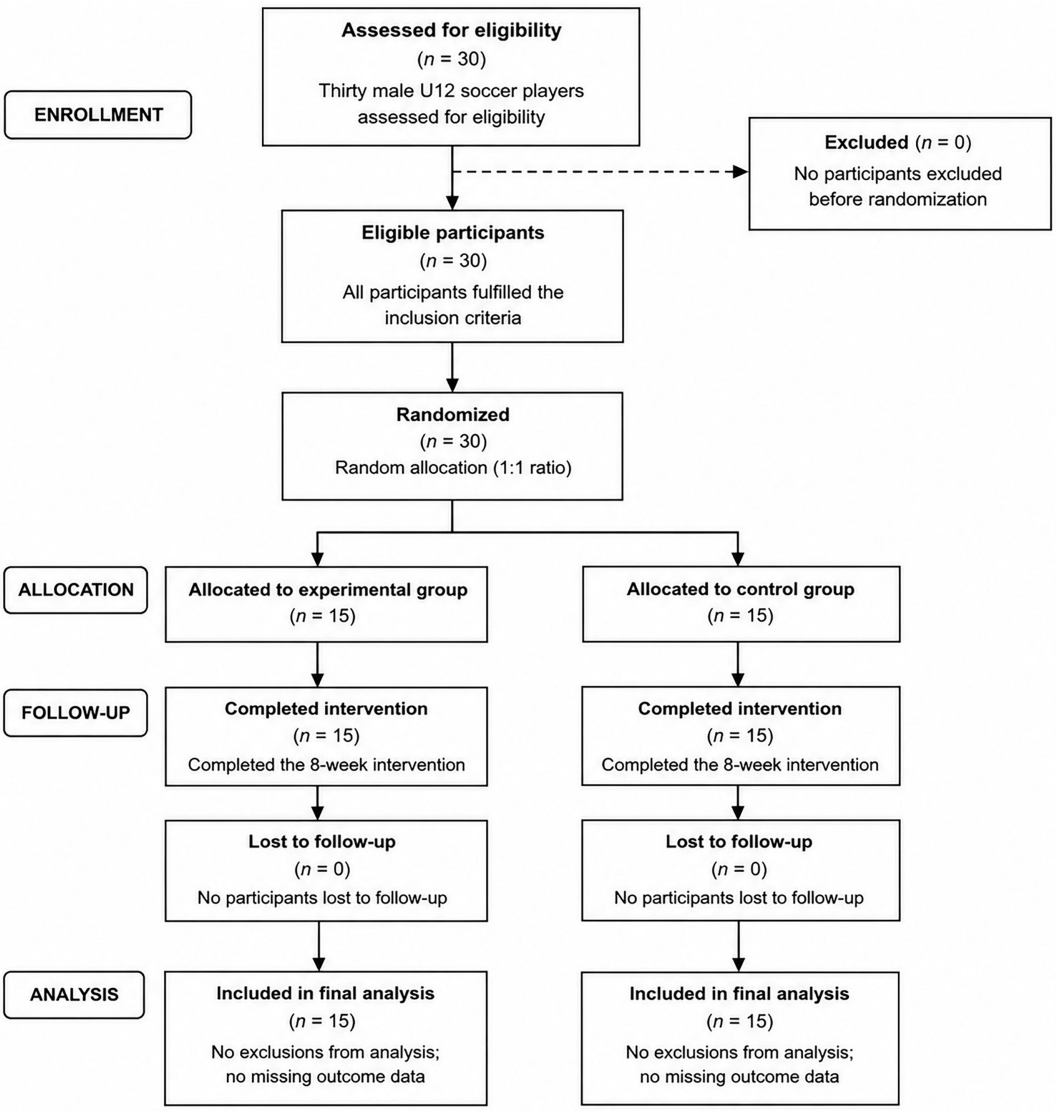


FIGURE 1
CONSORT flow diagram of participant progression through the study.

TABLE 2 Control group training program over the 8-week intervention period.

Weeks	Training component	Session content	Duration	Progression / task demands
1–2	Passing and ball control	Short- and medium-distance passing; oriented ball control	15 min	Basic and controlled technical execution
	Dribbling	Straight-line and change-of-direction dribbling	15 min	Controlled speed and simple movement patterns
	Shooting	Stationary and accuracy shooting	15 min	Emphasis on technical accuracy and controlled execution
	Integrated soccer activities	Small-sided games	15 min	Basic application of technical skills in game situations
3–4	Passing and ball control	Triangle passing; oriented control in movement	15 min	Increased movement and coordination demands
	Dribbling	Slalom dribbling	15 min	Increased directional-change demands
	Shooting	Shooting after passing or ball control	15 min	Integration of sequential technical actions
	Integrated soccer activities	Reduced-sided collective play	15 min	Increased decision-making and interaction demands
5–6	Passing and ball control	One-touch passing; passing in movement	15 min	Increased execution speed and coordination
	Dribbling	Dribbling with changes of pace or direction	15 min	Increased speed and movement complexity
	Shooting	Shooting after slalom or under pressure	15 min	Increased technical and situational demands
	Integrated soccer activities	Possession drills	15 min	Increased decision-making and game-specific demands
7–8	Passing and ball control	Quick short- and long-distance passing; dynamic control	15 min	High-speed technical execution
	Dribbling	Timed slalom	15 min	Increased speed and movement demands
	Shooting	Combined control-dribbling-shooting tasks	15 min	Integration of multiple technical actions
	Integrated soccer activities	Match-play application	15 min	Highest level of soccer-specific integration

informed of the participants' group assignments. Pre- and post-intervention assessments were conducted under similar standardized conditions between 16:00 and 18:00. Before each testing session, participants completed the same standardized soccer-specific warm-up routine. The Star Excursion Balance Test was performed barefoot, whereas soccer-specific skill tests were conducted using soccer shoes. The same footwear conditions were maintained during both assessment sessions. A one-minute recovery interval was provided between repeated trials for the applicable assessment procedures, whereas 2–3 min of recovery was provided between trials of the slalom running and dribbling tests. Participants were instructed to avoid strenuous physical activity for 24 h before each testing session. Participants were familiarized with all testing protocols before data collection. Evaluations were performed over four consecutive days under similar environmental conditions. On Day 1, anthropometric measurements were obtained. On Day 2, dynamic postural control was assessed using the Star Excursion Balance Test (SEBT). On Day 3, soccer-specific skill tests, including the Loughborough Soccer Shooting Test (LSST) and juggling test, were performed. On Day 4, agility and slalom dribbling tests were conducted.

2.3 Assessment procedures

All participants were evaluated before and after the 8-week intervention under the same testing conditions. The assessment battery included measures of dynamic postural and soccer-specific skill performance.

2.3.1 Anthropometric measurements

Height was measured using a portable stadiometer (Seca 216, Hamburg, Germany), and body mass was measured with the participants barefoot and wearing light clothing, using an EKSR Focus 9800 scale (Gislaved, Sweden). Body mass index (BMI, kg/m^2) was calculated as body mass (kg) divided by height squared (m^2). In addition, biological maturity was estimated using the maturity offset method, expressed as years from predicted peak height velocity (PHV). Years from PHV were predicted using the following equation proposed by Moore et al. (2015): Maturity offset = $-7.999994 + (0.0036124 \times \text{age} \times \text{height})$, where age is expressed in years and height is expressed in centimeters. A negative maturity

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TABLE 3 Experimental group training program over the 8-week intervention period.

Weeks	Training component	Session content	Duration	Progression / task demands
1–2	Non-dominant-leg passing	Short- and medium-distance passing with the non-dominant leg	15 min	Basic and controlled execution
	Non-dominant-leg dribbling	Straight-line and basic change-of-direction dribbling with the non-dominant leg	15 min	Controlled speed and movement patterns
	Non-dominant-leg shooting	Stationary and target shooting with the non-dominant leg	15 min	Emphasis on technical accuracy and controlled execution
	Dynamic postural control	Bipodal stance on balance cushions with ball-handling tasks; 5 × 1-min exercise bouts with 2-min recovery periods	15 min	Basic unstable-surface postural control
3–4	Non-dominant-leg passing	Triangle passing with the non-dominant leg	15 min	Increased coordination and movement demands
	Non-dominant-leg dribbling	Slalom dribbling with the non-dominant leg	15 min	Increased directional-change demands
	Non-dominant-leg shooting	Shooting after passing or ball control with the non-dominant leg	15 min	Integration of sequential technical actions
	Dynamic postural control	Bipodal stance on balance cushions with visual and hand-pass tasks; 5 × 1-min exercise bouts with 2-min recovery periods	15 min	Increased visual and coordination demands
5–6	Non-dominant-leg passing	One-touch passing with the non-dominant leg	15 min	Increased execution speed and coordination
	Non-dominant-leg dribbling	Dribbling with changes of pace or direction using the non-dominant leg	15 min	Increased speed and movement complexity
	Non-dominant-leg shooting	Target shooting with the non-dominant leg	15 min	Increased accuracy and technical demands
	Dynamic postural control	Bipodal stance on balance cushions with light perturbation and hand-based ball reception/return tasks; 5 × 1-min exercise bouts with 2-min recovery periods	15 min	Increased perturbation and coordination demands
7–8	Non-dominant-leg passing and control	Quick passing and dynamic control with the non-dominant leg	15 min	High-speed technical execution
	Non-dominant-leg dribbling	Timed slalom with the non-dominant leg	15 min	Increased speed and movement demands
	Non-dominant-leg combined skills	Combined control-dribbling-shooting tasks with the non-dominant leg	15 min	Highest technical and movement demands
	Dynamic postural control	Bipodal stance on balance cushions with visual and combined hand tasks; 5 × 1-min exercise bouts with 2-min recovery periods	15 min	Highest postural-control and coordination demands

Each training component lasted 15 min. The postural-control exercise consisted of five 1-min bouts with 2-min recovery periods.

offset indicates that the participant was estimated to be before predicted PHV, whereas a positive value indicates that the participant was estimated to be after predicted PHV. This estimate was used only to characterize the maturity status of the participants. Because maturity offset provides an indirect prediction rather than a direct measure of biological maturation, the estimates were interpreted cautiously. Prediction error may occur at the individual level, particularly among early- and late-maturing individuals. Therefore, maturity offset values were not considered a direct measurement of biological maturity.

2.3.2 Dynamic postural control

Dynamic postural control was assessed on the second testing day using the Star Excursion Balance Test (SEBT), as originally described by Hertel et al. (2000). This unilateral balance test combines single-leg stance with maximal reach of the contralateral leg. Participants stood at the center of a grid consisting of eight lines extending at 45° increments in the following directions: anterolateral, anterior, antero-medial, medial, posteromedial, posterior, posterolateral, and lateral. Before formal testing, participants completed three familiarization trials to become familiar with the SEBT procedure and minimize potential learning effects. The eight reach directions were assessed in a fixed clockwise sequence, and the order of testing was not randomized. The dominant leg was always tested first, followed by the non-dominant leg, and the same testing order was maintained during both pre- and post-intervention assessments. Three trials were completed in each direction, and a one-minute recovery interval was provided between repeated trials. A trial was considered invalid if the participant lost balance, moved the stance foot from the center position, or used the reaching leg to provide additional support through contact with the floor. Invalid trials were repeated according to the testing procedure. Reach distances were normalized to leg length using the following formula: (reach distance/leg length) × 100. The mean value of the three valid trials in each direction was retained for analysis. The reliability of the SEBT measurements was assessed using ICC values calculated from the present sample for each reach direction and limb. The ICC

values ranged from 0.775 to 0.970 for the dominant leg and from 0.645 to 0.970 for the non-dominant leg at pre-intervention, and from 0.775 to 0.970 and 0.829 to 0.941, respectively, at post-intervention.

2.3.3 Dynamic balance asymmetry

Dynamic balance asymmetry was calculated from the normalized reach distances obtained during the SEBT. Inter-limb asymmetry was determined by comparing the performance of the dominant and non-dominant legs in each reach direction. Asymmetry values were calculated using the following formula, as described by Bishop et al. (2018):

$$Asymmetry\,(\%) = dominant - non-dominant/dominant \times 100$$

Inter-limb asymmetry was calculated as an absolute percentage difference between dominant and non-dominant limb performance. The dominant limb was selected as the reference limb for calculation purposes. Higher asymmetry values indicate a greater magnitude of difference between limbs, without indicating the direction of the imbalance.

This approach allows the identification of potential inter-limb imbalances, which may provide information about differences in bilateral performance.

2.3.4 Soccer-specific skill tests

Soccer-specific skills were assessed before and after the 8-week intervention using four field-based tests administered in the following order: the Loughborough Soccer Shooting Test, the juggling test, a change of direction test, and the slalom dribble test. These tests were selected to assess complementary components of soccer performance in under-12 soccer players, including shooting performance, ball control, change-of-direction speed, and dribbling ability.

2.3.5 Loughborough soccer shooting test

Shooting performance was assessed using the Loughborough Soccer Shooting Test, adapted from the protocol described by Ali et al. (2007). The test area was marked on the floor using 5-cm-wide yellow tape. The 5.5-m line and the goalkeeper's area were identified according to the experimenter's instructions. The shooting zone consisted of a square area measuring 8.56 × 8.5 m, with the nearest line located 16.5 m from the goal line. Four tall traffic cones were placed at each corner of the shooting zone, and a standard gymnasium bench was positioned at the center of the far side of the zone to serve as a rebound board. Each participant completed three shooting trials, with each trial consisting of one shot performed with the dominant leg and one shot performed with the non-dominant leg. Thus, three attempts were completed for each leg. Shooting performance was evaluated according to predefined scoring zones of the goal, with points awarded according to the accuracy of shot placement. Higher scores indicated better shooting accuracy. The best score obtained for each leg across the three trials was retained for subsequent analysis. Invalid attempts, including shots performed outside the designated shooting area or not following the required testing procedure, were excluded from the analysis. The reliability of the Loughborough Soccer Shooting Test was assessed using ICC values calculated from the present sample. The ICC values were 0.760 for the dominant leg and 0.825 for the non-dominant leg at pre-intervention, and 0.668 and 0.823, respectively, at post-intervention.

2.3.6 Juggling test

Ball control ability was assessed using a juggling test performed within a restricted 3 × 3 m area, following the procedure described by Bekris and Gioldasis (2016). Players were instructed to keep the ball airborne while contacting it sequentially with predetermined body parts: foot, thigh, and head, repeating the sequence continuously throughout the trial. The ball was not allowed to touch the ground or leave the testing area. The juggling test was performed separately using the dominant and non-dominant legs. In each condition, participants were instructed to use only the assigned leg during foot contacts, and the same testing procedure was applied for both limbs. Each player performed three trials, and the score for each trial corresponded to the maximum number of successful consecutive juggles completed while respecting the required sequence. The best score across the three trials was retained for analysis. The reliability of the juggling test was assessed using ICC values calculated from the present sample. The ICC values were 0.906 for the dominant leg and 0.933 for the non-dominant leg at pre-intervention, and 0.916 and 0.959, respectively, at post-intervention.

2.3.7 Slalom running and dribbling test

Change-of-direction and dribbling performances were assessed using a photoelectric cell system (Microgate, Bolzano, Italy), according to the procedure described by Dardouri et al. (2014). Players performed a maximal 15-m slalom course around seven cones. In the first condition, players started with both feet aligned at the starting line and completed the test without a ball, sprinting 1.5 m to the first cone, slaloming at maximal speed around the remaining six cones spaced 2 m apart, and then sprinting 1.5 m to the finish line. In the second condition, players performed the same slalom course while dribbling the ball separately with the dominant and non-dominant leg. Participants were instructed to use only the assigned leg for ball contacts during each dribbling condition, and compliance with the required dribbling leg was visually monitored by the investigators. Each test was performed twice with 2–3 min of recovery between trials, and the mean completion time was retained for subsequent analysis. No time penalty was applied when a player contacted a cone. Trials in which the player lost control of the ball during dribbling were considered invalid and were repeated. The reliability of the slalom running and dribbling tests was assessed using ICC values calculated from the present sample. For the slalom dribbling test, the ICC values were 0.977 for the dominant leg and 0.984 for the non-dominant leg at pre-intervention, and 0.957 and 0.991, respectively, at post-intervention. For the slalom running test, the ICC values were 0.973 at pre-intervention and 0.942 at post-intervention.

2.3.8 Skill index, inter-limb difference, and asymmetry

The skill index was calculated as the ratio between slalom running test time and slalom dribble test time, as previously described by Mirkov et al. (2008). To assess limb-specific performance, the skill index was calculated separately for the dominant and non-dominant legs. The inter-limb difference was defined as the absolute difference between these two values:

$$Inter-limb\ difference = dominant\ leg\ skill\ index - non-dominant\ leg\ skill\ index$$

To further quantify between-limb imbalance, skill index asymmetry was also expressed as a percentage, using the dominant leg as the reference value (Bishop et al., 2018):

$$Asymmetry\ (\%) = dominant\ leg\ skill\ index - non-dominant\ leg\ skill\ index / dominant\ leg\ skill\ index \times 100$$

Higher asymmetry values indicate a greater magnitude of difference between limbs, without indicating the direction of the imbalance. To determine the direction of the observed asymmetry, the signed asymmetry index (ASI) was calculated as follows: ASI (%) = [(ND − D)/((ND + D)/2)] × 100, where ND represents the non-dominant limb value and D represents the dominant limb value. Positive and negative ASI values were interpreted according to whether the observed performance difference favored the non-dominant or dominant limb, respectively.

2.4 Training program

Both groups participated in an 8-week supervised training intervention consisting of two sessions per week. Each session lasted 90 min and comprised a 15-min warm-up, four 15-min training drills, three 3-min recovery periods, and a 6-min cool-down. Training frequency, session duration, total training volume, and supervision conditions were identical between groups. The control group performed conventional soccer technical training, including passing, ball control, dribbling, shooting, and small-sided games. No specific instructions regarding limb dominance were provided, allowing players to use either leg according to the requirements of each task. The organization and progression of the control training program are presented in Table 2. The experimental group followed the same overall training structure; however, part of the conventional technical training was replaced by three technical exercises targeting non-dominant-leg performance and one dynamic bipodal balance exercise performed on unstable balance cushions. The non-dominant-leg exercises consisted of passing, dribbling, and shooting tasks performed using the non-dominant leg. Participants were instructed to use the assigned non-dominant leg during the relevant technical actions. The specific exercise organization, duration, and progression are presented in Table 3. Training progression was achieved through gradual increases in task complexity, technical demands, coordination requirements, execution speed, and soccer-specific movement demands over the 8-week intervention. The technical exercises progressed from basic and controlled execution toward more complex and dynamic soccer-specific tasks. The balance exercise was progressively adapted to increase the demands placed on postural regulation, coordination, and movement control, as detailed in Table 3. The unstable-surface exercise was incorporated to increase postural-control demands and challenge neuromuscular regulation during training. Balance cushions were used to provide an unstable support condition requiring greater regulation of body position and postural stability. A bipodal stance was selected to provide a controlled and progressive introduction to unstable-surface training in young U12 soccer players. The exercise was intended to complement the non-dominant-leg technical exercises by providing an additional postural-control stimulus within the same training session. Each group was supervised by a qualified soccer coach. Although different coaches supervised the two groups, both coaches followed the same training frequency, session duration, overall session structure, and total training volume to ensure comparable training exposure between groups. Training intensity and progression were controlled through changes in task complexity, execution speed, coordination demands, and movement requirements, while maintaining equivalent overall training volume between groups. All participants completed the 16 scheduled training sessions during the intervention period, corresponding to 100% adherence. No participant was excluded because of absence or low adherence. No injuries or adverse events were reported during the intervention period, and no participant was excluded because of injury, illness, or other adverse events.

2.5 Statistical analysis

All statistical analyses were conducted using JASP software (version 0.95.4; University of Amsterdam, Amsterdam, Netherlands). An *a priori* sample size calculation was performed using G*Power software (version 3.1), based on an ANOVA: repeated measures, within–between interaction design. The calculation was based on dynamic postural control assessed using the Star Excursion Balance Test (SEBT) as the primary outcome, with an alpha level of 0.05, a statistical power of 90%, and an expected medium effect size (f = 0.25). The analysis indicated that a minimum total sample size of 30 participants was required. Accordingly, 30 eligible male U12 soccer players were recruited and randomly allocated to the experimental group (n = 15) and control group (n = 15). All randomized participants completed the 8-week intervention and were included in the final analysis. Data distribution was assessed using the Shapiro–Wilk test, and homogeneity of variances was evaluated using Levene's test. Descriptive statistics are presented as mean ± standard deviation (SD). Baseline differences between groups were examined using independent-samples t-tests. Dynamic postural control assessed using the SEBT was considered the primary outcome, whereas soccer-specific skill performance (shooting, juggling, slalom running, and slalom dribbling), skill indices, inter-limb differences, and asymmetry indices were considered secondary outcomes. Both absolute and signed asymmetry indices were analyzed to quantify the magnitude and direction of inter-limb differences. Primary intervention effects were examined using two-way mixed analysis of variance (ANOVA) models, with group (experimental vs control) as a between-subject factor and time (pre-

intervention vs post-intervention) as a within-subject factor. These analyses were used to evaluate intervention-related changes over time and differences between groups for SEBT performance, inter-limb asymmetry measures, soccer-specific skill variables, and skill indices. Effect sizes for ANOVA models were reported as partial eta squared (η_p^2).

For outcomes involving repeated measurements of dominant and non-dominant limbs, three-way mixed ANOVA models were conducted, including group as a between-subject factor and time and limb as within-subject factors. These analyses were performed to determine whether intervention-related changes differed according to limb condition and to identify possible Group × Time × Limb interactions. For dynamic postural control assessed using the SEBT, Group × Time × Limb interactions were examined separately for each reach direction. For soccer-specific skill outcomes, dominant- and non-dominant-limb performances were analyzed according to limb-specific conditions.

Significant Group × Time or Group × Time × Limb interactions were further explored using Holm-adjusted pairwise comparisons to identify the specific conditions contributing to the observed effects. Mean differences (MD), 95% confidence intervals (CI), adjusted p-values (p_{Holm}), and Cohen's d effect sizes were reported for *post hoc* comparisons.

Inter-limb asymmetry was assessed using both absolute and signed asymmetry indices. Absolute asymmetry indices were used to quantify the magnitude of differences between dominant and non-dominant limbs, whereas signed asymmetry indices were used to determine the direction of these differences. For SEBT outcomes, signed asymmetry was examined separately for each reach direction, whereas for skill-related outcomes, signed asymmetry was calculated from skill index values. Positive and negative signed asymmetry values were interpreted according to the direction of the dominant–non-dominant limb difference and the characteristics of each outcome measure. For reliability analyses, single-measure intraclass correlation coefficients (ICC) were calculated using a two-way mixed-effects model with a fixed set of raters and absolute agreement, reported as $ICC_{(3,1)}$ with 95% confidence intervals. Statistical significance was set at $p < 0.05$."

3 Results

3.1 Dynamic postural control

Dynamic postural control outcomes following the 8-week intervention are presented separately for the dominant and non-dominant limbs (Tables 4 and 5). For the dominant limb, significant Group × Time interactions were observed across all eight SEBT reach directions. The experimental group demonstrated greater improvements in normalized reach distance compared with the control group across all tested directions. The magnitude of interaction effects varied from small to large, with the largest effects observed for the lateral, posterior, and anterior directions (Table 4). For the non-dominant limb, significant Group × Time interactions were also observed across all eight SEBT reach directions (Table 5). The experimental group generally demonstrated greater improvements compared with the control group, with the largest intervention-related effects observed for the lateral, posterolateral, and anterolateral directions. Although significant improvements were observed across most directions, the pattern of adaptation was not completely uniform. In particular, the anteromedial

TABLE 4 Effects of the 8-week intervention on mean ± standard deviation (SD) dominant-leg dynamic postural control across SEBT reach directions.

Variable	Group	Pre mean ± SD	Post mean ± SD	Δ %	Group×Time			Post hoc (Holm-adjusted)			
					F	*p*	ηp^2	contrast	95% CI	*p* Holm	Cohen's d
Anterior	Experimental	82.38 ± 4.50	84.91 ± 4.21	3.07	16.58	<0.001	0.37	CG vs EG	-2.34 to -0.64	< .001	-1.49
	Control	84.52 ± 1.59	84.69 ± 2.06	0.20							
Anteromedial	Experimental	89.25 ± 4.43	90.10 ± 3.62	0.95	4.46	0.044	0.14	CG vs EG	-1.52 to -0.02	.044	-0.77
	Control	91.21± 2.65	90.53 ± 3.25	0.75							
Medial	Experimental	90.22 ± 3.49	92.15 ± 3.16	2.14	10.62	0.003	0.28	CG vs EG	-1.96 to -0.42	.003	-1.19
	Control	90.60 ± 2.61	90.07± 2.81	0.59							
Posteromedial	Experimental	84.03± 7.74	86.88± 5.71	3.39	7.98	0.009	0.22	CG vs EG	-1.80 to -0.26	.009	-1.03
	Control	85.94 ± 8.26	85.29 ± 8.31	0.76							
Posterior	Experimental	66.51 ± 5.60	72.47± 4.98	8.96	17.82	<0.001	0.39	CG vs EG	-2.33 to -0.76	< .001	-1.54
	Control	70.51 ± 9.79	68.79 ± 7.47	2.44							
Posterolateral	Experimental	77.17 ± 3.94	81.17± 3.99	5.18	8.41	0.007	0.23	CG vs EG	-1.86 to -0.26	.007	-1.06
	Control	78.66 ± 4.00	78.93 ± 4.95	0.34							
Lateral	Experimental	75.89 ± 5.25	79.92 ± 5.26	5.31	27.16	<0.001	0.49	CG vs EG	-2.70 to -1.11	< .001	-1.90
	Control	78.79 ± 3.93	77.55 ± 4.05	1.57							
Anterolateral	Experimental	81.78 ± 4.46	83.81 ± 5.08	2.48	5.27	0.029	0.16	CG vs EG	-1.61 to -0.07	.029	-0.84
	Control	83.66 ± 4.07	83.35 ± 4.91	0.37							

TABLE 5 Effects of the 8-week intervention on mean ± standard deviation (SD) non-dominant-leg dynamic postural control across SEBT reach directions.

Variable	Group	Pre mean ± SD	Post mean ± SD	Δ %	Group×Time			Post hoc (Holm-adjusted)			
					F	*p*	η_p^2	contrast	95% CI	*p* Holm	Cohen's d
Anterior	Experimental	68.03 ± 4.82	72.18 ± 4.21	6.10	31.90	<0.001	0.533	CG vs EG	-2.91 to -1.21	< .001	-2.062
	Control	69.56 ± 4.15	68.86 ± 4.46	1.01							
Anteromedial	Experimental	89.52 ± 4.60	87.49 ± 4.40	2.27	14.36	<0.001	0.339	CG vs EG	0.61 to 2.16	< .001	1.384
	Control	88.09 ± 2.82	88.65± 3.13	0.64							
Medial	Experimental	92.80 ± 3.65	93.06± 3.62	0.28	3.36	0.018	0.185	CG vs EG	-1.69 to -0.16	.018	-0.921
	Control	91.72 ± 2.71	90.46 ± 2.72	1.37							
Posteromedial	Experimental	90.83 ± 6.32	91.93 ± 4.62	1.21	6.16	0.019	0.180	CG vs EG	-1.66 to -0.15	.019	-0.907
	Control	92.68 ± 2.93	89.57 ± 6.27	3.36							
Posterior	Experimental	78.89 ± 4.82	81.99 ± 3.99	3.93	17.24	<0.001	0.381	CG vs EG	-2.37 to -0.66	< .001	-1.516
	Control	81. 50 ± 4.01	81.48 ± 4.52	0.02							
Posterolateral	Experimental	82.57 ± 2.72	85.06 ± 2.57	3.02	35.81	<0.001	0.561	CG vs EG	-2.96 to -1.41	< .001	-2.185
	Control	84.22 ± 2.29	82.95 ± 2.66	1.51							
Lateral	Experimental	74.48 ± 4.01	78.60 ± 4.63	5.53	61.75	<0.001	0.688	CG vs EG	-3.72 to -2.02	< .001	-2.869
	Control	75.96 ± 2.57	74.64 ± 3.09	1.74							
Anterolateral	Experimental	76.40 ± 5.29	78.86 ± 5.37	3.22	29.98	<0.001	0.517	CG vs EG	-2.77 to -1.23	< .001	-1.999
	Control	77.08 ± 4.07	75.94 ± 3.93	1.48							

direction showed a different pattern of change, with a decrease in normalized reach distance in the experimental group and a slight increase in the control group. Nevertheless, the significant Group × Time interaction indicated a different trajectory of change between groups. Overall, these findings indicate that the 8-week intervention improved dynamic postural control in both dominant and non-dominant limbs, although the magnitude of adaptation varied according to SEBT reach direction.

3.2 Limb-specific effects on dynamic postural control

To determine whether intervention-related changes in dynamic postural control differed according to limb condition, three-way mixed ANOVA models including group, time, and limb were performed (Table 6). The Group × Time × Limb interaction was examined separately for each SEBT reach direction. A significant three-way interaction was observed only for the anteromedial direction ($F = 4.49$, $p = 0.037$, $\eta_p^2 = 0.05$). No significant Group × Time × Limb interactions were observed for the remaining SEBT directions, including anterior, medial, posteromedial, posterior, posterolateral, lateral, and anterolateral directions (all $p > 0.05$). The significant anteromedial interaction was further examined using Holm-adjusted *post hoc* comparisons. A significant pre–post improvement was observed for the non-dominant limb in the experimental group (Mean Difference = 2.04, 95% CI = 0.37 to 3.71, $p_{Holm} = 0.007$, Cohen's d = 0.55) (Table 6). These findings suggest that intervention-related adaptations were direction-specific and that differences between dominant and non-dominant limbs were limited to specific movement conditions rather than representing a generalized bilateral effect.

TABLE 6 Three-way mixed ANOVA results and post hoc comparisons following Group × Time × Limb interactions for dynamic postural control.

SEBT direction	F (Group × Time × Limb)	p-value	η_p^2	Post hoc (Holm-adjusted)
Anterior	1.216	0.273	0.014	—
Anteromedial	4.493	0.037*	0.051	MD = 2.037; 95% CI = 0.365 to 3.709; $_pHolm = 0.007$; d = 0.552
Medial	0.426	0.516	0.005	—
Posteromedial	0.062	0.805	0.001	—
Posterior	1.593	0.210	0.019	—
Posterolateral	0.000	0.983	<0.001	—
Lateral	0.005	0.945	<0.001	—
Anterolateral	0.189	0.665	0.002	—

TABLE 7 Effects of the 8-week intervention on mean ± standard deviation (SD) inter-limb asymmetry in dynamic postural control.

Variable	Group	Pre mean ± SD	Post mean ± SD	Δ %	Group×Time			Post hoc (Holm-adjusted)			
					F	p	η_p^2	contrast	95% CI	p Holm	Cohen's d
Anterior	Experimental	17.17 ± 7.43	14.83 ± 5.87	13.63	8.50	0.007	0.233	CG vs EG	0.308 to 1.821	.007	1.065
	Control	17.69 ± 5.09	18.71 ± 4.48	5.77							
Anteromedial	Experimental	4.67± 3.66	3.24± 2.96	30.64	0.752	0.393	0.026	CG vs EG	0.436 to 1.070	.393	0.317
	Control	3.77 ± 3.58	3.10 ± 3.90	17.77							
Medial	Experimental	3.58 ± 2.78	2.30 ± 2.10	35.75	0.525	0.475	0.018	CG vs EG	-0.487 to 1.016	.475	0.264
	Control	3.29 ± 1.42	2.68 ± 2.71	18.54							
Posteromedial	Experimental	9.11 ± 8.18	6.31 ± 4.25	30.74	0.273	0.605	0.010	CG vs EG	-0.559 to 0.941	.605	0.191
	Control	8.87± 7.52	6.98 ± 8.01	21.31							
Posterior	Experimental	19.66 ± 12.63	14.59 ± 8.36	25.79	8.054	0.008	0.223	CG vs EG	0.277 to 1.795	.008	1.036
	Control	19.31 ± 9.79	21.21 ± 9.68	9.84							
Posterolateral	Experimental	7.13 ± 3.93	5.10 ± 3.65	28.47	0.048	0.829	0.002	CG vs EG	-0.669 to 0.828	.829	0.080
	Control	7.23 ± 4.10	5.61 ± 6.11	22.41							
Lateral	Experimental	6.49 ± 4.43	4.52 ± 3.27	30.35	2.88	0.101	0.093	CG vs EG	-0.143 to 1.382	.101	0.620
	Control	5.49± 3.42	5.61 ± 4.87	2.19							
Anterolateral	Experimental	8.03± 7.02	6.99 ± 6.72	12.95	1.604	0.216	0.054	CG vs EG	- 0.287 to 1.212	.216	0.462
	Control	8.21 ± 5.82	8.26 ± 6.61	0.61							

3.3 Inter-limb asymmetry and signed asymmetry in dynamic postural control

Changes in inter-limb asymmetry and the direction of asymmetry across SEBT reach directions are presented in Tables 7 and 8. For absolute inter-limb asymmetry, significant Group × Time interactions were observed for the anterior and posterior directions (Table 7). In the anterior direction, asymmetry decreased in the experimental group from 17.2 ± 7.4% to 14.8 ± 5.9%, whereas it increased in the control group from 17.7 ± 5.1% to 18.7 ± 4.5%. Similarly, in the posterior direction, asymmetry decreased in the experimental group from 19.7 ± 12.6% to 14.6 ± 8.4%, while it increased in the control group from 19.3 ± 9.8% to 21.2 ± 9.7%. No significant Group × Time interactions were observed for the

TABLE 8 Directional changes in signed asymmetry index for dynamic postural control (SEBT).

Direction	Group	Pre ASI (%) Mean ± SD	Post ASI (%) Mean ± SD	F	p	η_p^2	Post hoc (Holm-adjusted)
Anterior	Experimental	-19.13 ± 9.09	-16.22 ± 6.84	8.229	0.008*	0.227	MD=-2.914; 95% CI -5.795 to -0.033; $_p$Holm=0.046; d=-0.419
	Control	-19.56 ± 6.04	-20.76 ± 5.29				
Anteromedial	Experimental	0.30 ± 5.35	-2.98 ± 5.34	16.819	<0.001*	0.375	MD=3.278; 95% CI 0.989 to 5.567; $_p$Holm=0.002; d=0.659
	Control	-3.49 ± 4.44	-2.09 ± 4.70				
Medial	Experimental	2.82 ± 3.42	0.96 ± 2.95	1.096	0.304	0.038	—
	Control	1.23 ± 3.44	0.43 ± 3.92				
Posteromedial	Experimental	7.91 ± 7.93	5.72 ± 4.46	0.090	0.766	0.003	—
	Control	7.93 ± 7.18	5.10 ± 8.25				
Posterior	Experimental	17.14 ± 10.93	12.40 ± 8.85	6.879	0.014*	0.197	MD=4.740; 95% CI -0.478 to 9.959; $_p$Holm=0.093; d=0.435
	Control	15.12 ± 12.11	17.20 ± 11.42				
Posterolateral	Experimental	6.83 ± 3.63	4.75 ± 3.71	0.022	0.883	<0.001	—
	Control	6.92 ± 3.80	5.10 ± 5.85				
Lateral	Experimental	-1.78 ± 7.89	-1.63 ± 5.56	0.050	0.825	0.002	—
	Control	-3.60 ± 5.81	-3.78 ± 6.98				
Anterolateral	Experimental	-6.87 ± 9.56	-6.12 ± 8.73	1.838	0.186	0.062	—
	Control	-8.20 ± 7.31	-9.26 ± 7.58				

*$_p$ < 0.05.

TABLE 9 Effects of the 8-week intervention on mean ± standard deviation (SD) dominant-leg soccer-specific skill performance.

Variable	Group	Pre mean ± SD	Post mean ± SD	Δ %	Group×Time			Post hoc (Holm-adjusted)			
					F	p	η_p^2	contrast	95% CI	p Holm	Cohen's d
Shooting at the Goal (Number)	Experimental	3.86 ± 0.64	4.60 ± 0.50	19.17	0.45	0.508	0.016	CG vs EG	-1.00 to 0.51	0.508	-0.245
	Control	4.20 ± 0.77	4.73 ± 0.45	12.62							
Juggling test (Number)	Experimental	19.00 ± 2.13	22.47 ± 2.58	18.26	0.46	0.504	0.016	CG vs EG	-0.504 to 1.00	0.504	0.247
	Control	20.27 ± 1.87	24.13 ±1.24	19.04							
Slalom Running Test	Experimental	8.28 ± 0.72	7.46 ± 0.54	9.90	4.86	0.036	0.148	CG vs EG	0.03 to 1.59	0.036	0.806
	Control	8.32 ± 0.63	7.72 ± 0.57	7.21							
Slalom Dribble Test	Experimental	11.79 ± 1.11	9.93 ± 0.71	15.78	2.24	0.145	0.074	CG vs EG	-0.22 to 1.31	0.145	0.547
	Control	11.99 ± 1.10	10.45 ± 0.96	12.84							

anteromedial, medial, posteromedial, posterolateral, lateral, or anterolateral directions. Although reductions in asymmetry magnitude were observed in several directions in the experimental group, these changes did not consistently differ from those observed in the control group. The signed asymmetry index analysis provided additional information regarding the direction of inter-limb differences (Table 8). Significant Group × Time interactions were observed for the anterior and anteromedial directions. In the anterior direction, the experimental group showed a shift in signed asymmetry values from -19.1 ± 9.1% to -16.2 ± 6.8%, whereas the control group changed from -19.6 ± 6.0% to -20.8 ± 5.3%. For the anteromedial direction, the experimental group changed from 0.30 ± 5.4% to -3.0 ± 5.3%, while the control group changed from -3.5 ± 4.4% to -2.1 ± 4.7%. Overall, these findings indicate that the intervention influenced both the magnitude and direction of inter-limb differences in specific SEBT directions. The combination of absolute and signed asymmetry analyses provided complementary information regarding bilateral adaptations.

3.4 Soccer-specific skill performance

The effects of the 8-week intervention on soccer-specific skill performance are presented separately for dominant- and non-dominant-limb conditions (Tables 9, 10). For dominant-limb performance, a significant Group × Time interaction was observed only for the Slalom Running Test. No significant Group × Time interactions were observed for dominant-limb shooting performance, juggling performance, or slalom dribbling performance (Table 9). For non-dominant-limb performance, significant Group × Time interactions were observed for shooting, juggling, and slalom dribbling performance (Table 10). The experimental group demonstrated marked improvements in non-dominant-limb shooting performance, whereas no comparable improvement was observed in the control group. Similarly, non-dominant-limb juggling performance increased substantially following the intervention in the experimental group, while only minimal changes were observed in the control group. For non-dominant-limb slalom dribbling performance, completion time decreased in the experimental group, indicating improved performance, whereas changes in the control group were limited. These findings indicate that the intervention produced greater improvements in soccer-specific skill performance, particularly for non-dominant-limb conditions, suggesting enhanced bilateral skill development.

3.5 Limb-specific effects on soccer-specific skill performance

To further determine whether intervention-related changes in soccer-specific performance differed according to limb condition, three-way mixed ANOVA models including group, time, and limb

TABLE 10 Effects of the 8-week intervention on mean ± standard deviation (SD) non-dominant-leg soccer-specific skill performance.

Variable	Group	Pre mean ± SD	Post mean ± SD	Δ %	Group×Time			Post hoc (Holm-adjusted)			
					F	p	η_p^2	contrast	95% CI	p Holm	Cohen's d
Shooting at the Goal (LSST)	Experimental	1.73 ± 1.03	3.80 ± 0.67	119.65	29.00	0.001	0.509	CG vs EG	-2.76 to -1.17	< .001	-1.966
	Control	1.73 ± 1.10	1.73 ± 0.96	0							
juggling test (Number)	Experimental	9.13 ± 2.50	15.80 ± 2.45	73.05	192.0	0.001	0.873	CG vs EG	-6.63 to -3.49	< .001	-5.060
	Control	8.20 ± 2.80	8.40 ± 2.72	2.44							
Slalom Running Test (s)	Experimental	8.28 ± 0.72	7.46 ± 0.54	9.90	4.86	0.036	0.148	CG vs EG	0.03 to 1.59	0.036	0.806
	Control	8.32 ± 0.63	7.72 ± 0.57	7.21							
Slalom Dribble Test (s)	Experimental	15.71 ± 1.15	12.58 ± 0.88	19.92	159.1	.001	0.850	CG vs EG	3.14 to 6.07	< .001	4.606
	Control	15.58 ± 0.90	15.09 ± 0.89	3.15							

TABLE 11 Three-way mixed ANOVA results for soccer-specific performance including limb dominance.

Variable	F (Group × Time × Limb)	p-value	η_p^2
Shooting (LSST)	13.40	<0.001	0.131
Juggling test	26.65	<0.001	0.221
Slalom dribbling test	43.62	<0.001	0.308
Skill Index	15.46	<0.001	0.148

were conducted (Table 11). Significant Group × Time × Limb interactions were observed for shooting performance, juggling performance, slalom dribbling performance, and Skill Index (Table 11). These findings indicate that intervention-related changes differ according to limb condition. The significant interactions were further explored using Holm-adjusted *post hoc* comparisons (Table 12). Significant pre–post improvements were observed in the experimental group across the evaluated soccer-specific skills, including both dominant- and non-dominant-limb conditions. The *post hoc* analyses confirmed that the intervention-induced improvements were greater than those observed in the control group. Overall, these findings indicate that the intervention promoted limb-specific adaptations and improved bilateral soccer-specific skill performance rather than producing changes restricted to the dominant limb.

3.6 Skill indices and inter-limb asymmetry indices

The effects of the 8-week intervention on skill indices, inter-limb differences, and asymmetry indices are presented in Table 13. For the dominant-limb skill index, no significant Group × Time interaction was observed, indicating that changes in dominant-limb skill index did not differ between groups. In contrast, a significant Group × Time interaction was observed for the non-dominant-limb skill index. The experimental group demonstrated an increase in skill index values from 0.52 ± 0.04 to 0.59 ± 0.05, whereas the control group showed a decrease from 0.53 ± 0.04 to 0.51 ± 0.04. The Holm-adjusted *post hoc* comparison confirmed a significant between-group difference in change scores with a large effect size (Table 13). A significant Group × Time interaction was also observed for inter-limb difference. The experimental group showed a reduction in inter-limb difference values, whereas the control group demonstrated an increase following the intervention. Similarly, the absolute asymmetry index showed a significant Group × Time interaction. The experimental group demonstrated a reduction in asymmetry from 25.0 ± 3.2% to 20.9 ± 5.9%, whereas the control group showed an increase from 23.0 ± 4.9% to 30.7 ± 4.3%. These findings indicate a reduction in the magnitude of inter-limb imbalance following the intervention. The signed asymmetry index also showed a significant Group × Time interaction. The experimental group demonstrated a shift in signed asymmetry values from -28.6 ± 4.2% to -23.5 ± 7.4%, whereas the control group showed a further decrease from -26.2 ± 6.2% to -36.4 ± 6.0%. These results indicate that the intervention affected not only the magnitude of asymmetry but also its direction. Overall, the combined analysis of skill indices, absolute asymmetry, and signed asymmetry indices suggests that the intervention particularly enhanced non-dominant-limb performance and promoted a more balanced bilateral skill profile.

4 Discussion

The present study investigated the effects of an 8-week combined training program incorporating non-dominant-limb-focused technical exercises and unstable-surface balance exercises on dynamic postural control, soccer-specific skill performance, and inter-limb asymmetry in male U12 soccer players. The experimental group demonstrated greater

TABLE 12 Post hoc comparisons following Group × Time × Limb interactions for soccer-specific performance.

Outcome	Post hoc contrast	Mean Difference	95% CI	*p* Holm	Cohen's d
Shooting Test	Experimental: Pre Dominant vs Post Dominant	-0.733	-1.461 to -0.006	0.018	-0.915
	Experimental: Pre NonDominant vs Post NonDominant	-2.067	-3.272 to -0.861	<0.001	-2.580
Juggling Test	Experimental: Pre Dominant vs Post Dominant	-3.467	-4.910 to -2.023	<0.001	-2.400
	Experimental: Pre NonDominant vs Post NonDominant	-6.667	-7.806 to -5.528	<0.001	-4.246
Slalom Dribbling Test	Experimental: Pre Dominant vs Post Dominant	1.863	1.337 to 2.390	<0.001	1.907
	Experimental: Pre NonDominant vs Post NonDominant	3.123	2.614 to 3.633	<0.001	3.197
Skill Index	Experimental: Pre Dominant vs Post Dominant	-0.047	-0.086 to -0.007	0.003	-0.805
	Experimental: Pre NonDominant vs Post NonDominant	-0.066	-0.089 to -0.044	<0.001	-1.147

Post hoc comparisons were performed following significant Group × Time × Limb interactions. p-values and confidence intervals were adjusted for multiple comparisons using Holm correction. Cohen's d values indicate the direction of the pre–post contrast; the magnitude of the effect was interpreted using the absolute value of Cohen's d.

TABLE 13 Effects of the 8-week intervention on mean ± standard deviation (SD) skill indices and inter-limb asymmetry indices.

Variable	Group	Pre mean ± SD	Post mean ± SD	Δ %	Group×Time			Post hoc (Holm-adjusted)			
					F	p	η_p^2	contrast	95% CI	p Holm	Cohen's d
Dominant Index	Experimental	0.70 ± 0.06	0.75 ± 0.04	7.14	0.02	0.87	0.001	CG vs EG	-0.81 to 0.69	.876	-0.058
	Control	0.69 ± 0.08	0.74 ± 0.07	7.25							
Non-dominant Index	Experimental	0.52 ± 0.04	0.59 ± 0.05	13.46	93.07	<0.001	0.769	CG vs EG	-4.41 to -2.64	< .001	-3.523
	Control	0.53 ± 0.04	0.51 ± 0.04	3.77							
Inter-limb difference	Experimental	0.17 ± 0.03	0.15 ± 0.04	11.76	24.62	<0.001	0.468	CG vs EG	-1.75 to -0.16	.014	-0.953
	Control	0.16 ± 0.04	0.23 ± 0.05	43.75							
Asymmetry index (%)	Experimental	24.98 ± 3.20	20.88 ± 5.92	16.41	41.27	<0.001	0.596	CG vs EG	1.57 to 3.12	< .001	2.346
	Control	23.03 ± 4.86	30.72 ± 4.34	33.39							
Signed asymmetry index (%)	Experimental	-28.61 ± 4.20	-23.54 ± 7.36	17.72	42.40	<0.001	0.602	CG vs EG	-19.85 to -5.96	<0.001	-2.141
	Control	-26.18 ± 6.16	-36.44 ± 5.97	-39.18							

improvements than the control group in several dynamic postural control and soccer-specific skill outcomes. However, the magnitude and direction of adaptations varied according to the assessed task, movement direction, and limb condition. The analyses considering limb dominance and signed asymmetry indices further indicated that intervention-related adaptations were not uniformly distributed between dominant and non-dominant limbs.

The improvements observed in SEBT performance are consistent with previous evidence showing that neuromuscular and balance-oriented training can enhance dynamic postural control in young soccer players (Jlid et al., 2019; Mitrousis et al., 2023). The combined intervention may have provided complementary stimuli through unstable-surface exercises and repeated unilateral technical actions. However, these adaptations should not be interpreted as evidence of direct transfer from bilateral balance tasks to unilateral technical performance. Instead, unstable-surface exercises may have contributed to general postural-control and sensorimotor capacities that support sport-specific movements, whereas improvements in non-dominant-limb technical performance likely resulted primarily from repeated practice of non-dominant-limb tasks. Because both training components were applied simultaneously, their independent contributions cannot be determined. Therefore, the observed adaptations should be interpreted as responses to the combined training program. The changes observed in SEBT performance should also be interpreted cautiously. The SEBT provides information regarding functional dynamic postural control and lower-limb coordination during challenging unilateral tasks; however, it does not represent all components of postural control and should not be interpreted as a direct measure of soccer-specific performance. Therefore, changes in SEBT outcomes should be considered as complementary functional adaptations rather than direct evidence of improved soccer ability.

The effects on dynamic postural control were direction-specific. Although improvements were observed across several SEBT directions for both limbs, the Group × Time × Limb analysis revealed a significant limb-dependent effect only for the anteromedial direction. This finding suggests that training adaptations were influenced by the specific mechanical and coordinative requirements of each movement direction rather than reflecting a generalized difference between dominant and non-dominant limbs. Since SEBT performance requires the integration of strength, balance, joint mobility, and neuromuscular coordination, variations between directions are expected.

The intervention produced greater improvements in soccer-specific skill performance, particularly under non-dominant-limb conditions. Significant improvements in non-dominant shooting, juggling, and slalom dribbling suggest that repeated exposure to non-dominant-limb tasks may enhance technical proficiency and bilateral skill development. These findings are consistent with previous research showing that non-preferred-foot training can improve technical abilities in young soccer players (Guilherme et al., 2015). Nevertheless, the present design does not allow the specific contribution of non-dominant-limb training to be separated from the effects of balance training, maturation, familiarization, and general training adaptations.

The analysis of skill indices provided additional insight into bilateral performance changes. The intervention improved non-dominant-limb skill index values and reduced inter-limb differences and asymmetry indices, suggesting a more balanced bilateral performance profile. These findings indicate that the intervention may contribute not only to improving performance but also to reducing disparities between limbs. However, these adaptations should not be interpreted as direct evidence of inter-limb transfer, as repeated exposure to testing procedures and general training effects may also contribute to performance improvements.

Inter-limb asymmetry was evaluated using both absolute and signed asymmetry indices. Absolute asymmetry provides information about the magnitude of differences between limbs, whereas signed asymmetry identifies the direction of these differences. Previous research has highlighted that asymmetry in athletes is influenced by the specific task, performance variable, and methodological approach used for assessment (Bishop et al., 2018; Maloney, 2019). The present findings showed that asymmetry patterns varied according to the evaluated task and movement direction, supporting the concept that asymmetry is a context-dependent characteristic rather than a universal marker of reduced performance.

The combined assessment of absolute and signed asymmetry indices represents an important methodological aspect of the

present study. While reductions in absolute asymmetry indicate changes in the magnitude of bilateral differences, signed asymmetry provides additional information regarding the direction of these changes. Consequently, both measures should be considered when monitoring bilateral adaptations in young athletes, as the functional relevance of asymmetry may depend on the specific task, movement demands, and performance variable.

Biological maturation should also be considered when interpreting training responses in U12 soccer players. Differences in growth status, neuromuscular control, strength development, coordination, and musculoskeletal characteristics may influence individual responsiveness to training during youth development (Malina et al., 2004, 2015). Although maturity offset was estimated in the present study, this approach provides only an indirect indication of biological maturation. Future studies should include more comprehensive maturation assessments to better characterize training adaptations during youth development.

4.1 Limitations

Several limitations should be acknowledged. First, although the sample size was determined using an *a priori* power analysis, the relatively small sample of male U12 soccer players may limit the generalizability of the findings to other populations, including female players, older athletes, or different competitive levels. Second, the intervention combined non-dominant-limb technical training and unstable-surface balance exercises; therefore, the independent contribution of each component cannot be determined. Future studies using separate intervention conditions are required to clarify the specific and combined effects of these training approaches. Third, the study relied on field-based assessments. Although SEBT and soccer-specific skill tests provide ecologically relevant measures, they do not capture all neuromuscular mechanisms underlying performance adaptations. Furthermore, the use of a field-based dynamic postural control assessment represents a limitation of the present study. Although the SEBT is widely applied in athletic populations, it reflects functional reach performance under specific conditions and does not provide a complete assessment of all mechanisms involved in postural control. In addition, asymmetry indices, including signed asymmetry, remain dependent on the characteristics of the assessed task and should be interpreted within the specific performance context. Although signed asymmetry analysis improves interpretation by identifying the direction of inter-limb differences, it does not determine whether a given asymmetry pattern is functionally advantageous or disadvantageous. Finally, although the number of outcomes increased the number of statistical comparisons, Holm-adjusted corrections were applied for *post hoc* analyses to reduce the risk of type I error. Familiarization procedures were implemented; however, repeated exposure to testing tasks may have contributed partially to observed improvements. The absence of long-term follow-up also prevents conclusions regarding the persistence of the observed adaptations.

4.2 Practical implications

The findings provide preliminary support for incorporating combined technical exercises emphasizing the non-dominant limb together with unstable-surface balance activities into training programs for young soccer players. This approach may contribute to bilateral technical development while providing additional sensorimotor challenges that may contribute to improved dynamic postural control. The assessment of both absolute and signed asymmetry indices may provide coaches and practitioners with a more comprehensive method for monitoring bilateral adaptations by identifying both the magnitude and direction of inter-limb differences. However, these implications should remain specific to the studied population and training context. The present findings do not allow determining whether non-dominant-limb training or unstable-surface balance training independently contributed to the observed adaptations. Future research using larger samples, separate intervention groups, detailed maturation assessments, and longer follow-up periods is required to confirm these findings and clarify the contribution of each training component.

5 Conclusion

An 8-week combined training program incorporating non-dominant-limb technical exercises and unstable-surface balance activities was associated with greater improvements than the control condition in dynamic postural control, selected non-dominant-limb soccer-specific skill outcomes, and bilateral performance balance, including inter-limb asymmetry outcomes, in male U12 soccer players. The intervention particularly enhanced non-dominant-limb skill performance and reduced inter-limb differences, while absolute and signed asymmetry analyses provided complementary information regarding the magnitude and direction of bilateral adaptations. However, the relatively small sample size, the inclusion of multiple outcomes, and the absence of separate intervention conditions limit the ability to determine the independent contribution of each training component. Therefore, these findings should be interpreted as preliminary and specific to male U12 soccer players. Future studies with larger samples, clearly defined primary outcomes, separate intervention groups, comprehensive maturation assessment, and longer follow-up periods are needed to confirm these findings and clarify the mechanisms underlying training-induced adaptations.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

The studies involving humans were approved by The Ethics Committee of the Deanship of Scientific Research at Palestine Technical University–Kadoorie. The studies were conducted in accordance with the local legislation and institutional requirements. Written informed consent for participation in this study was

provided by the participants' legal guardians/next of kin. Written informed consent was obtained from the individual(s), and minor (s)' legal guardian/next of kin, for the publication of any potentially identifiable images or data included in this article.

Author contributions

HAM: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Writing – original draft. TP: Validation, Writing – review & editing. RvdT: Conceptualization, Methodology, Validation, Visualization, Writing – review & editing. JC: Conceptualization, Methodology, Validation, Visualization, Writing – review & editing. JA: Visualization, Writing – review & editing. MCJ: Conceptualization, Methodology, Project administration, Validation, Writing – review & editing.

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Conflict of interest

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